

Package ‘ButterflyLS’

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Type Package

Title Butterfly Longevity Estimation

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Description Provides methods and data to enable reproduction of the results presented in ``Estimation of adult butterfly longevity using long-term citizen-science count data" (Clarke, Dennis and McCrea 2026)

License GPL (>= 2)

Depends R(>= 3.5),dplyr,reshape2

LazyData true

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NeedsCompilation no

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analysis

TODO - Brief description

Description

TODO - Longer description here (maybe one or two sentences)

Usage

```
analysis(data, year, setup_level)
```

Arguments

data	dataframe containing the data to analysis
year	character value corresponding to the the year to analyse from the data
setup_level	character value "daily" or "weekly" dependent on how the data was sampled.

Value

TODO describe the ouput structure here

Examples

```
library(ButterflyLS)

# lazy load data
data(dark_green_fritillary_weekly)
# year to analyse
year <- "2018"
# weekly sampling
setup_level <- "weekly"

# run analysis
results <- analysis(dark_green_fritillary_weekly,year,setup_level)
```

analysis_multiyear*TODO - Brief description*

Description

TODO - Longer description here (maybe one or two sentences)

Usage

```
analysis_multiyear(data, setup_level, phi_type)
```

Arguments

data	dataframe containing the data to analysis
setup_level	character value "doubleslope", "slope", "variable" or "fixed"

Value

TODO describe the output structure here

Examples

```
library(ButterflyLS)

# lazy load data
data(orange_tip_weekly)
# year to analyse
year <- "2018"
# weekly sampling
setup_level <- "weekly"
# phi type
phi_type = "slope"

# run analysis
results <- analysis_multiyear(orange_tip_weekly, setup_level, phi_type)
```

dark_green_fritillary_daily
<i>Dark Green Fritillary</i>

Description

Daily site, year, and count data for dark green fritillary

Usage

```
data(dark_green_fritillary_daily )
```

Format

A dataframe with 5 columns and 581129 rows. The columns are DAYNO (day number), WEE-KNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLS)

data(dark_green_fritillary_daily )
head(dark_green_fritillary_daily )
plot(dark_green_fritillary_daily $COUNT)
```

```
dark_green_fritillary_weekly
```

Dark Green Fritillary Weekly

Description

Weekly site, year, and count data for dark green fritillary

Usage

```
data(dark_green_fritillary_weekly)
```

Format

A dataframe with 5 columns and 82420 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLS)

data(dark_green_fritillary_weekly)
head(dark_green_fritillary_weekly)
plot(dark_green_fritillary_weekly$COUNT)
```

```
marbled_white_daily
```

Marbled White

Description

Daily site, year, and count data for marbled white

Usage

```
data(marbled_white_daily)
```

Format

A dataframe with 5 columns and 1048575 rows. The columns are DAYNO (day number), WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLS)

data(marbled_white_daily)
head(marbled_white_daily)
plot(marbled_white_daily$COUNT)
```

marbled_white_weekly	<i>Marbled White Weekly</i>
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Description

Weekly site, year, and count data for marbled white

Usage

```
data(marbled_white_weekly)
```

Format

A dataframe with 5 columns and 206778 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLS)

data(marbled_white_weekly)
head(marbled_white_weekly)
plot(marbled_white_weekly$COUNT)
```

orange_tip_daily	<i>Orange Tip Daily</i>
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Description

Daily site, year, and count data for orange tip

Usage

```
data(orange_tip_daily)
```

Format

A dataframe with 5 columns and 1048575 rows. The columns are DAYNO (day number), WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), COUNT (count), and DATE (date).

Examples

```
library(ButterflyLS)

data(orange_tip_daily)
head(orange_tip_daily)
plot(orange_tip_daily$COUNT)
```

orange_tip_weekly	<i>Orange Tip Weekly</i>
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Description

Weekly site, year, and count data for orange tip

Usage

```
data(orange_tip_weekly)
```

Format

A dataframe with 5 columns and 346164 rows. The columns are WEEKNO (week number), SITENO (site number), YEAR (year), SPECIES (species identify), and COUNT (count).

Examples

```
library(ButterflyLS)

data(orange_tip_weekly)
head(orange_tip_weekly)
plot(orange_tip_weekly$COUNT)
```

simulation	<i>TODO - Brief description</i>
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Description

TODO - Longer description here (maybe one or two sentences)

Usage

```
simulation(
  setup_level,
  env_cov.type,
  a.type,
  dist.type,
  B,
  cov_sim_type,
  n.iter,
  dpw,
  meth,
  convert_weekly,
  vary_phi,
  NA_type,
  nS,
  year
)
```

Arguments

setup_level	character value "daily" or "weekly" dependent on how the data is to be sampled.
env_cov.type	how to add environmental covariate data, either "lin", "quad" or "none"
a.type	"N", "SO" or "S"
dist.type	"P", "NB" or "ZIP"
B	number of broods
cov_sim_type	how covariate data is simulated - "uniform" or "seasonal"
n.iter	number of iterations
dpw	number of days per week to simulate

meth	optimisation method - "Nelder-Mead", "SANN" or "L-BFGS-B"
convert_weekly	convert from the daily to the weekly format - "yes" or "no"
vary_phi	vary phi over the season - "yes" or "no"
NA_type	missing values - "match_NA", "prop_NA" or "none"
nS	number of sites, if matching raw data then use the length of the output

Value

TODO describe the output structure here

Examples

```
library(ButterflyLS)

# setup simulation parameters
year <- "2018"
setup_level <- "daily"
env_cov.type <- "none"
a.type <- "S0"
dist.type <- "P"
B <- 1 ## single brood
cov_sim_type <- "uniform"
n.iter <- 250
dpw <- 1
meth <- "Nelder-Mead"
convert_weekly <- "no"
vary_phi <- "no"
NA_type <- "prop_NA"
nS <- 200

# run simulation
results <- simulation(setup_level,
  env_cov.type,
  a.type,
  dist.type,
  B,
  cov_sim_type,
  n.iter,
  dpw,
  meth,
  convert_weekly,
  vary_phi,
  NA_type,
  nS,
  year)
```

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